LIGHTWEIGHT ACOUSTIC LINING DESIGN FOR AIRCRAFT FUSELAGE

ME613 COMMERCIAL AIRPLANE GROUP – TEAM 4

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Design Characteristics

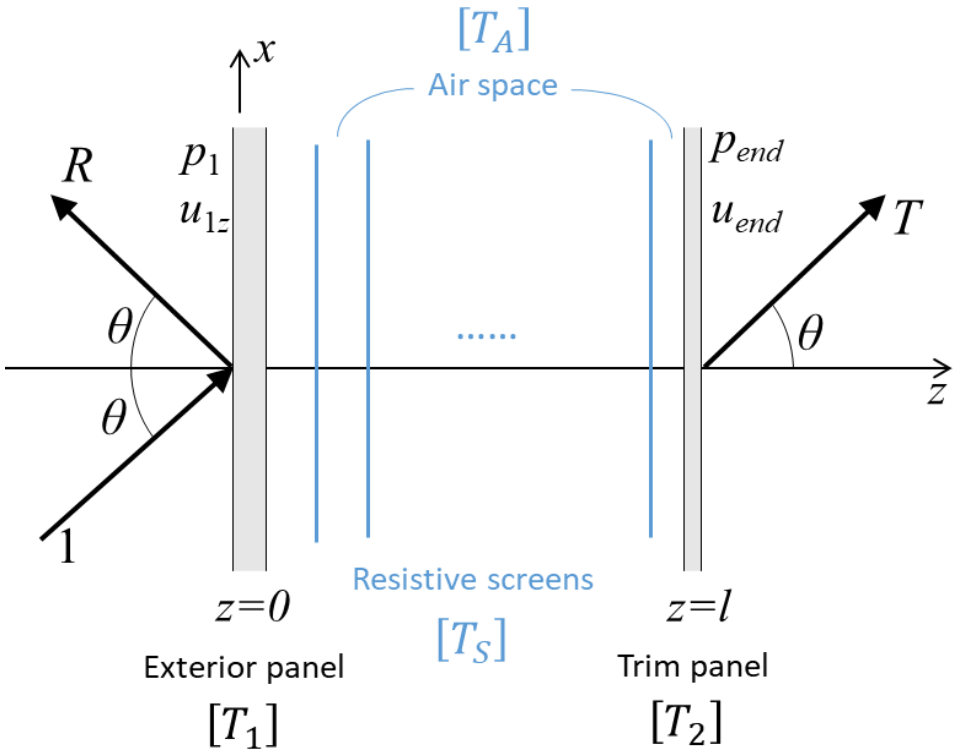
Goals + Constraints

- Frequency Analysis Range
  - 100-20k Hz
  - Emphasis on the Speech Interference Range (SIR) (500-4k Hz)
- Lightweight
  - Total combined m_s of all screens < 2 kg/m^2
- High Performance
  - High TL throughout the entire frequency range

Assumptions

- Thin Screens
  - Reasonable max of 100 screens could fit in the air gap
- Incompressible Flow
- Negligible Inertial Effects

System Illustration and Modeling



Transfer Matrix Modeling

For air space: $[T_A] = \begin{bmatrix} \cos(k_z l) & j\frac{\rho_0 c}{\cos\theta}\sin(k_z l) \\ j\frac{\cos\theta}{\rho_0 c}\sin(k_z l) & \cos(k_z l) \end{bmatrix}$

For panels: $[T] = \begin{bmatrix} 1 & z_m \\ 0 & 1 \end{bmatrix}$

Flexurally-stiff panel:

$$z_m = j(\omega m_s - \frac{D}{\omega}k_x^4) \quad D = \frac{Eh^3}{12(1-\upsilon^2)}(1+j\eta)$$

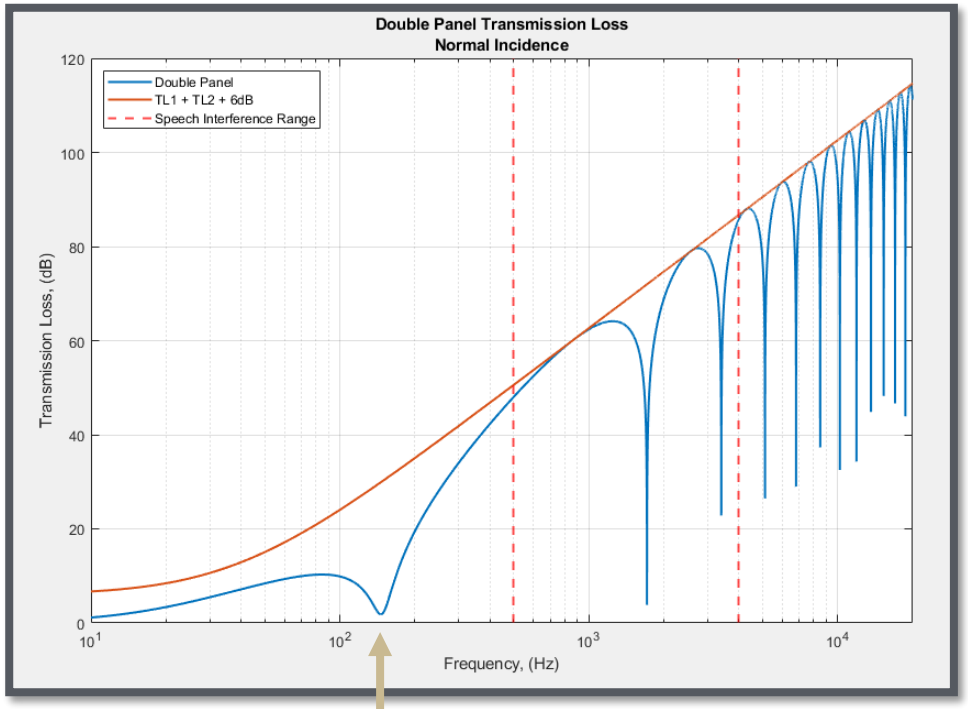
Flexible, resistive screen: $\frac{1}{z_m} = \frac{1}{R_f} + \frac{1}{j\omega m_s}$

$$\begin{bmatrix} p_1 \\ u_{1z} \end{bmatrix}_{z=0} = [T_1][T_{A1}][T_{S1}][T_{A2}][T_{S2}]\dots[T_{Sn}][T_{An}][T_2]\begin{bmatrix} p_{end} \\ u_{end} \end{bmatrix}_{z=l}$$

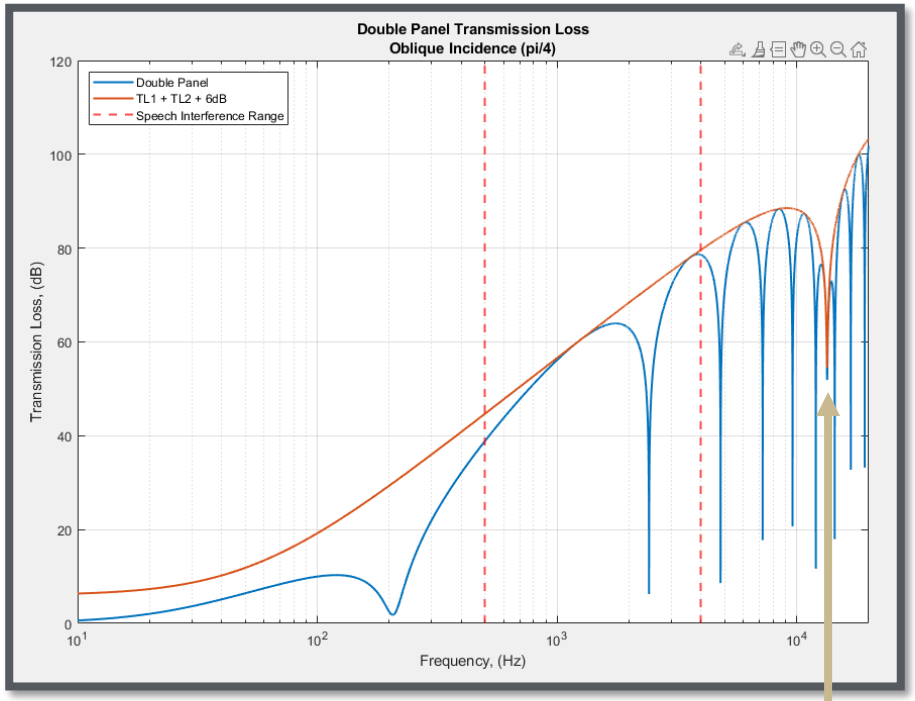
E: Young's modulus
η: loss factor
υ: Possion Ratio
h: thickness of the panel
R_f: flow resistence

Part 1 - i) Unlined Double Panel System

Double Panel System: Normal and Oblique Incidence



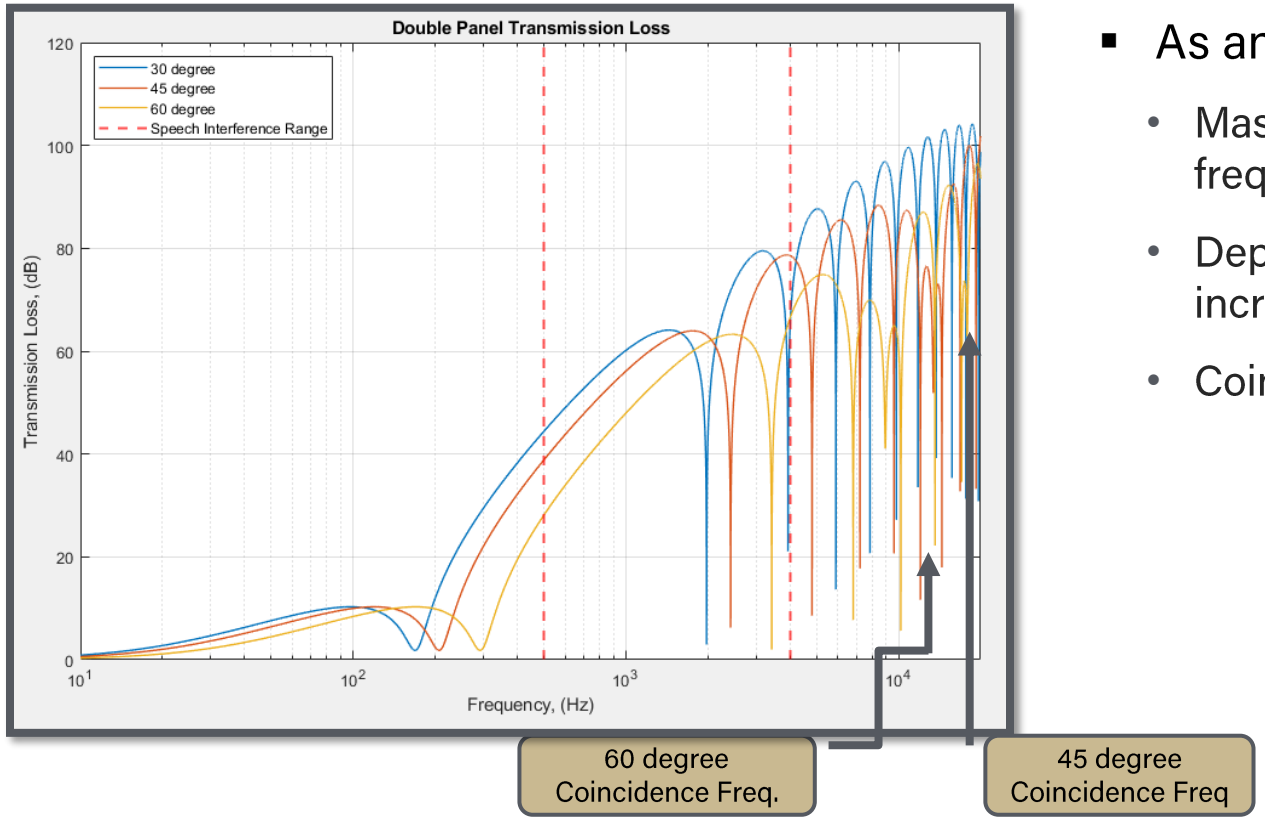
M-A-M Resonance



Coincidence Frequency

Part 1 - i) Unlined Double Panel System, cont.

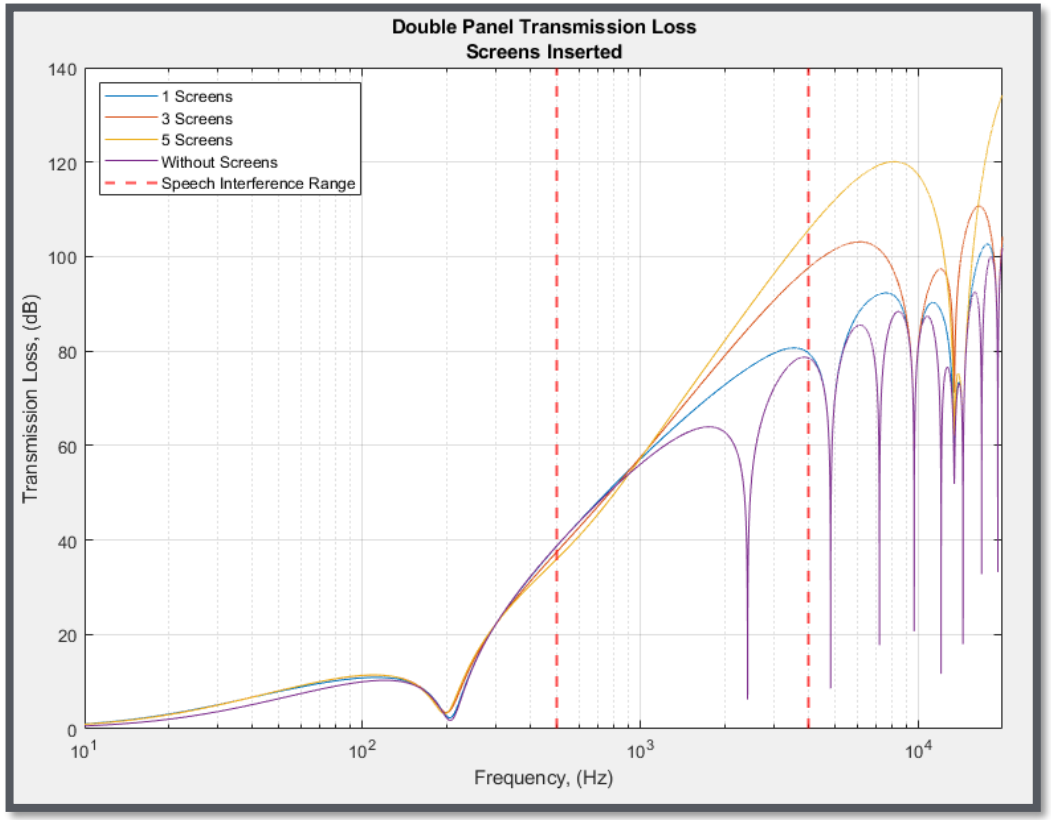
Double Panel System: Variation with Angle of Incidence



- As angle of incidence increases
  - Mass-Air-Mass resonance frequency increases
  - Depth resonance frequency increases
  - Coincidence frequency decreases

Part 1 - ii) Insertion of Flexible Resistive Screens

Insertion of various number of screens into system



Arbitrarily Chosen Values:

$$R_f = 1000\ (Rayls)$$
$$m_s = 2\ (\mathrm{kg/m^2})\ \text{in total}$$

- With insertion of screens
  - Mass-Air-Mass resonance and coincidence frequency changes little.
  - Depth resonances are eliminated.
  - For a constant total mass, the more screens inserted, the higher the transmission loss is.

Part 1 - iii) The Solution, Parameter Selection

Using identical screens

Number of Screens

Decoupling Frequency of Screens:

$$\omega_d \approx 2\pi f_d \approx \frac{R_f}{m_s}$$

$$f_d = 500 \text{ Hz}$$

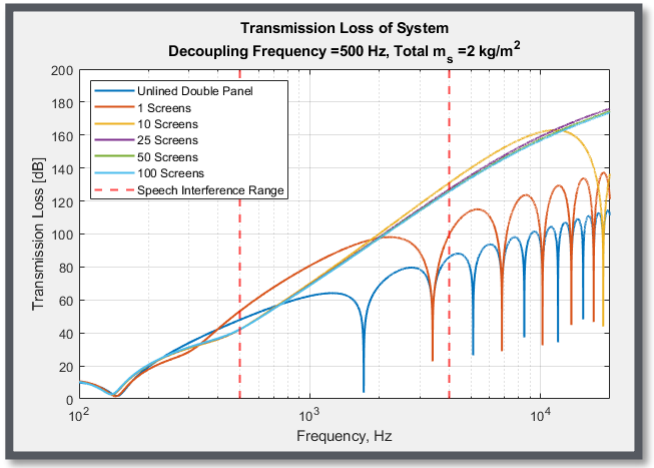
Maximize Total Mass of Screens:

$$m_{s_t} = 2 \Rightarrow m_s = \frac{2}{n}\frac{kg}{m^2}$$

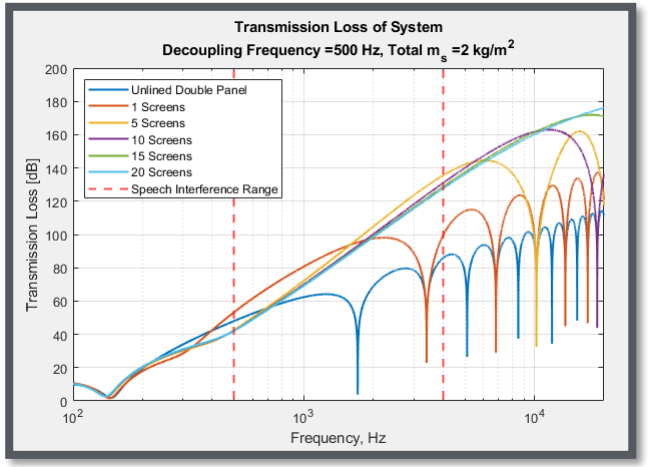
n= Number of Screens

Flow Resistance:

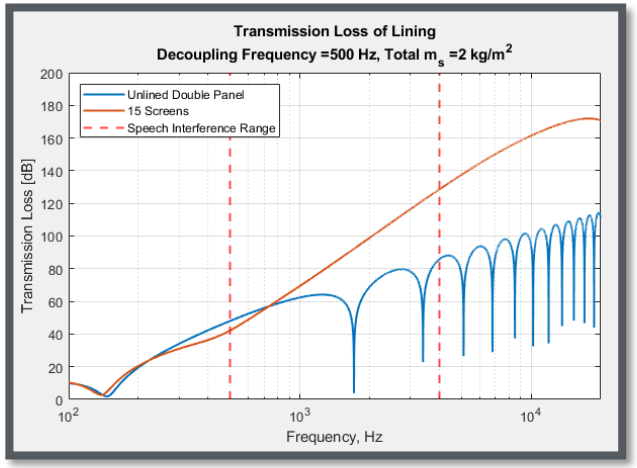
$$R_f = 2\pi f_d m_s$$



Wide Range of Screens



Convergence Test to Avoid High Frequency Depth Resonances

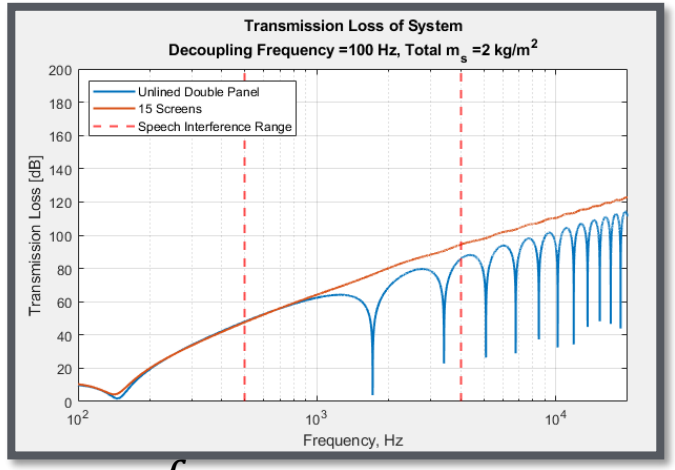


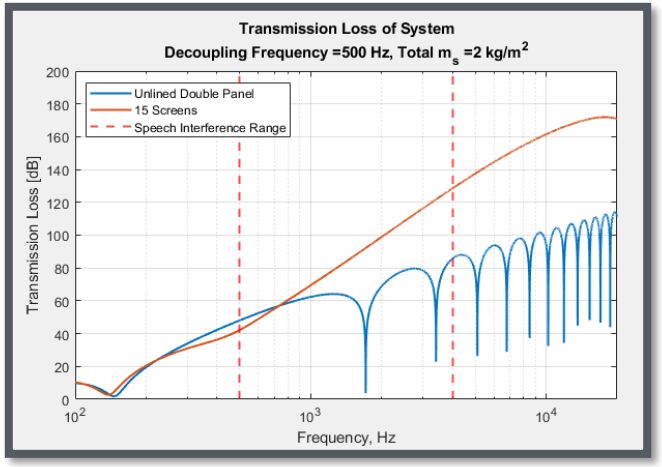
Minimum Number of Screens to Avoid High Frequency Depth Resonances

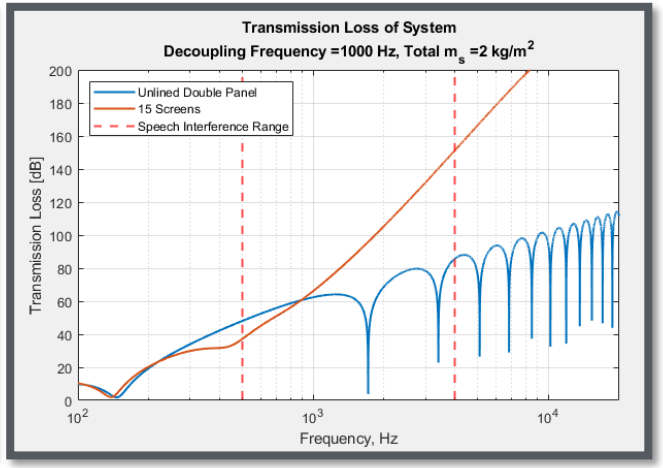
Part 1 - iii) The Solution, Parameter Selection, cont.

$$For\ 15\ Screens:\ m_s = \frac{2}{15} = .133\ \frac{kg}{m^2}$$

Decoupling Frequency







$$f_d/_5 = 100\ Hz$$

$$R_f = 2\pi\frac{f_d}{5}m_s \approx 84\ \text{Rayls}$$

$$f_d = 500\ \text{Hz}$$

$$R_f = 2\pi f_d m_s \approx 419\ \text{Rayls}$$

$$2f_d = 1000\ Hz$$

$$R_f = 2\pi(2f_d)m_s \approx 838\ \text{Rayls}$$

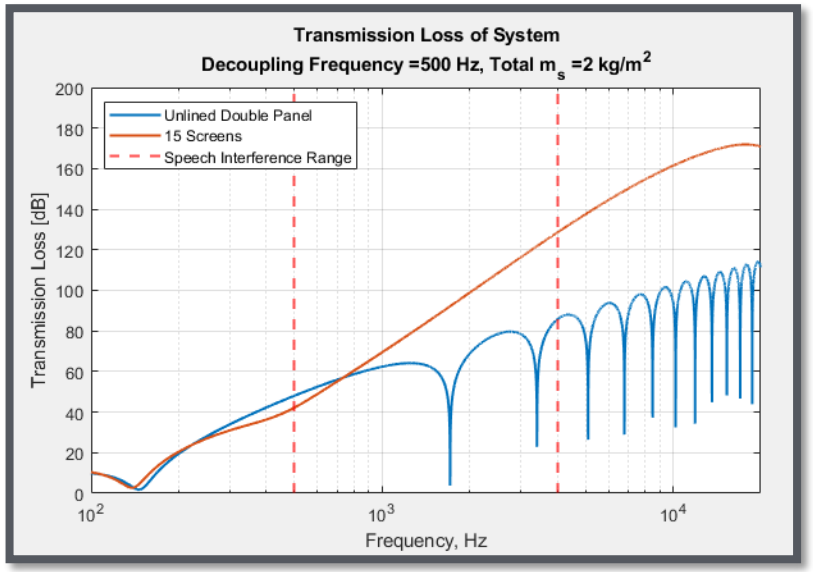
As f_d decreases, R_f decreases, and the system response approaches the unlined double panel

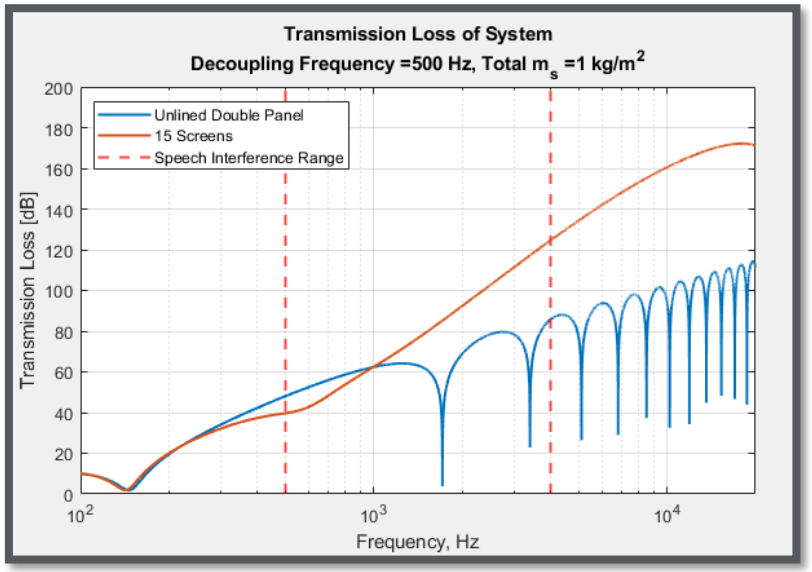
As f_d increases, R_f increases, and TL decreases at low frequencies, but increases at high frequencies

Part 1 - iii) The Solution, Parameter Selection, cont.

Verification of Mass Assumption

$Holding\ R_f, n\ constant$



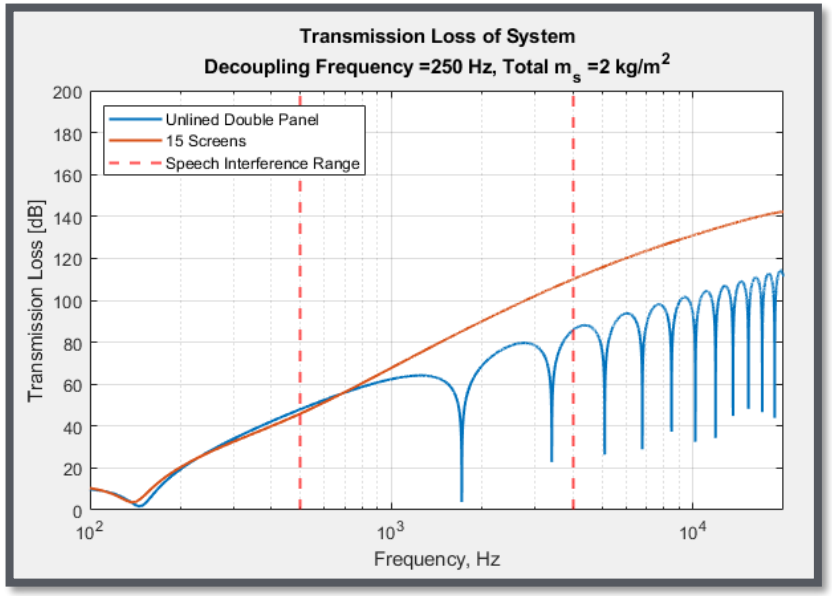


By decreasing the Mass we see decreased Transmission Loss in low frequency range

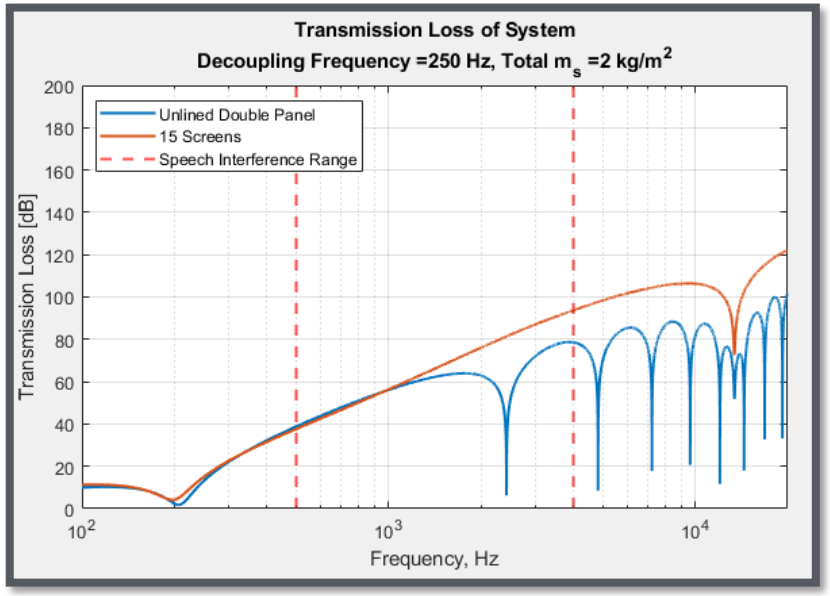
Therefore, the choice of using the maximum permitted Mass has been validated

Part 1 - iii) The Solution, Result

Normal Incidence



Oblique Incidence (π/4)



Final Tuned Parameters:

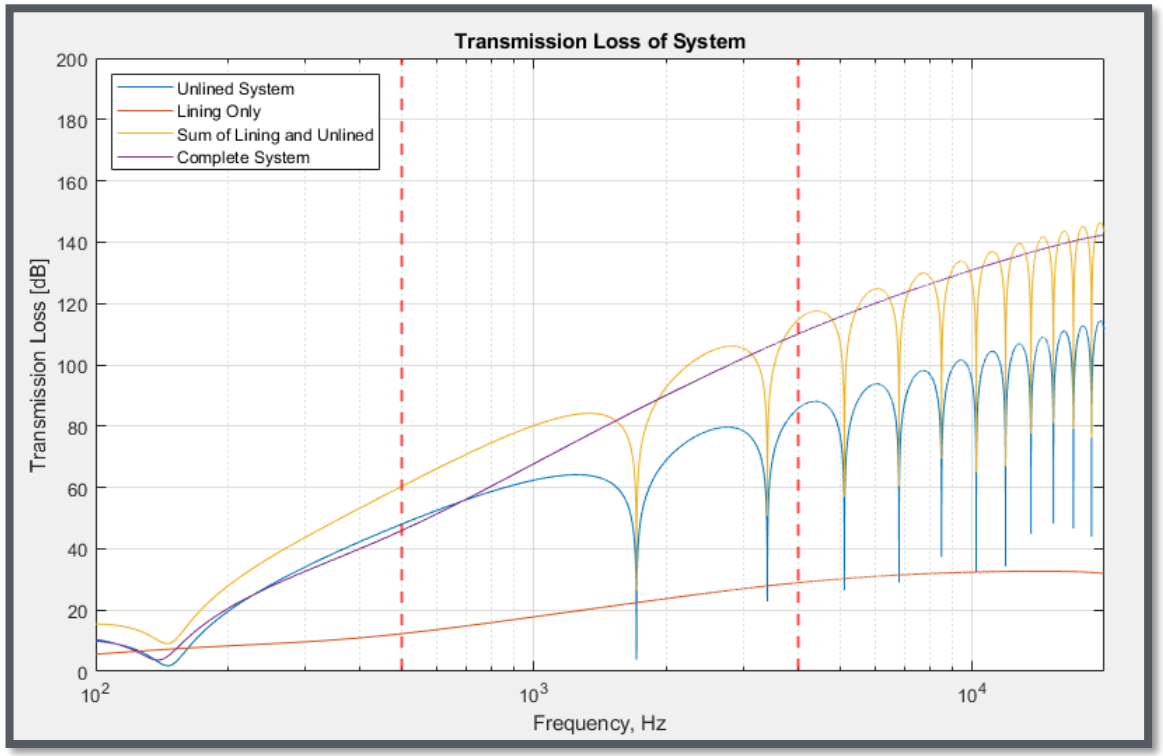
$$15\ Screens, m_s \approx .13\ \frac{kg}{m^2}$$

$$f_d = 250\ Hz, R_f \approx 209\ Rayls$$

- Observations
  - Slight decrease in TL in the frequency region slightly greater than M-A-M
    - Possible result of introduction of M-A-M Resonances between Screens
  - Coincidence Frequency does still appear for oblique incidence

Part 1 - iv) The Lining

Transmission Loss of Lining + Comparisons



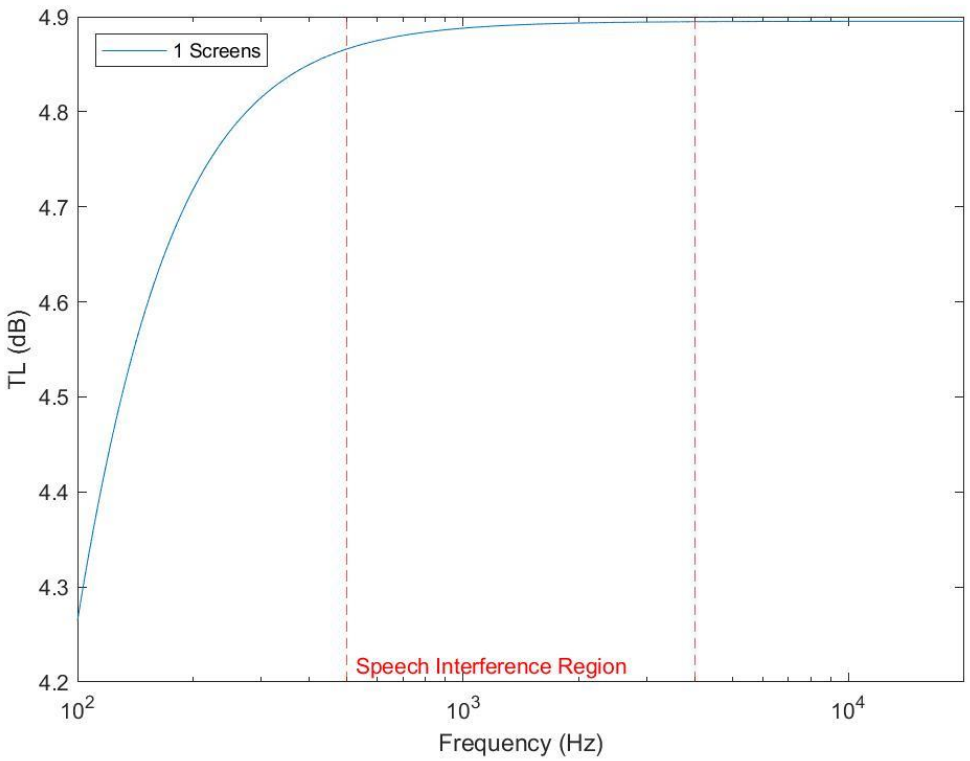
- Lining Conclusions
  - Peaks of sum of the lining and unlined system TL is greater than actual system TL
  - Depth resonances are eliminated in the target frequency range

Part 2 – Transmission Through a Single Screen

- Choose Rf to ensure "rigid" behavior (flat TL) in the Speech Interference Region
- Decoupling frequency:

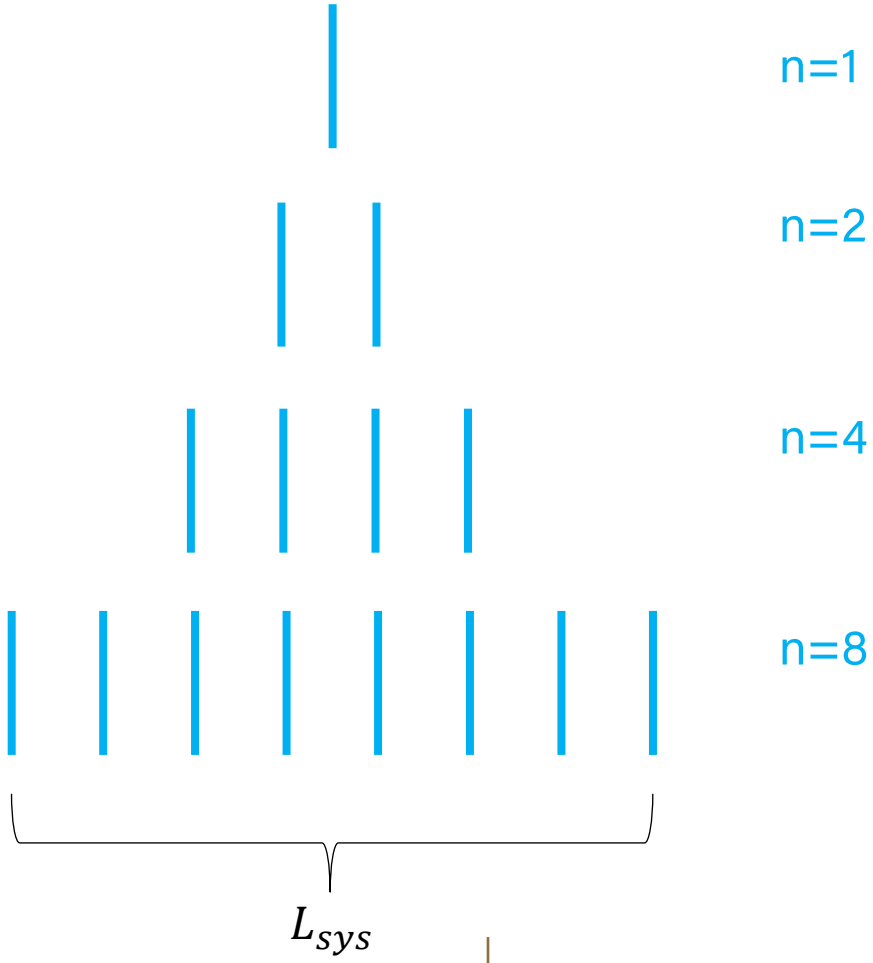
$$2\pi f_d \approx \frac{R_f}{m_s}$$

- Screen acts "rigid" when $f \gg f_d$
- Choose $f_d = 50\ \mathrm{Hz}$
- Thus: $m_s = 2\mathrm{kg}$, $R_f \approx 628\ \mathrm{Rayls}$
- TL approaches 4.895 dB limit



Part 2 - Convergence of Halving the Screens

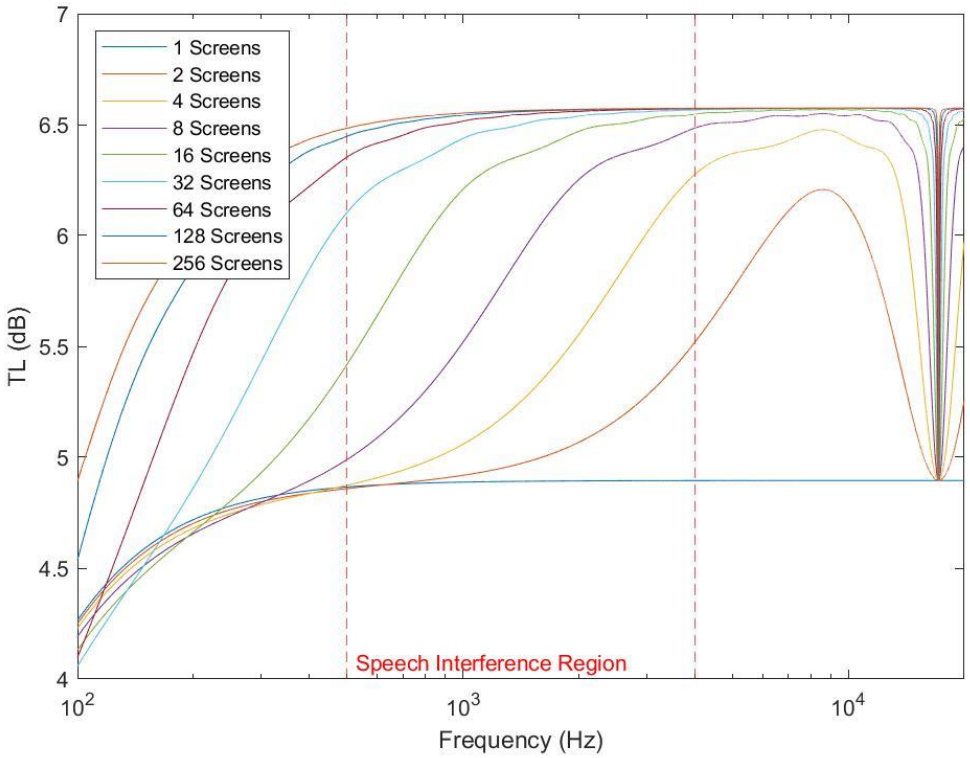
- Split mass/unit area:
  $$m_s = (2\ \mathrm{kg/m^2})/n$$
- Split flow resistance:
  $$R_f = (628\ \mathrm{Rayls})/n$$
- ω_d remains constant
- Increase total system length:
  $$L_{sys} = (1\ \mathrm{cm}) * (n-1)$$
- System transfer matrix:
  $$[T_{sys}] = ([T_s][T_A])^n[T_s]$$



Part 2 - Convergence of Halving the Screens

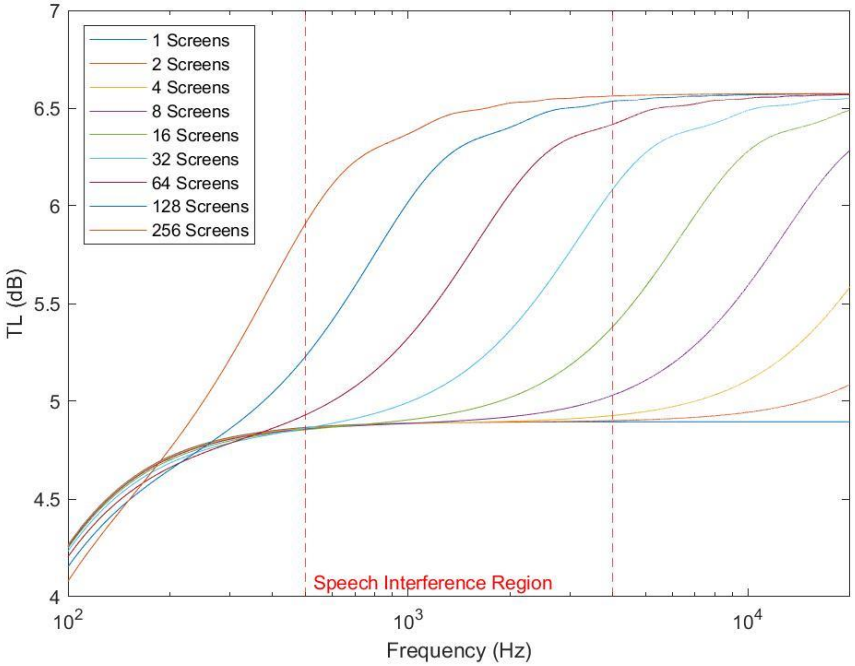
- Depth resonances appear for multiple screens, TL dips to the single screen line
- TL follows single screen then transitions to a 6.573 dB limit as n increases
- Depth resonances become narrower as n increases
- After sufficient halving operations, system behaves like a single rigid resistive screen in Speech Interference Region but at a higher TL and with depth resonances

1 cm air gap between screens



Part 2 – Air Space Depth Investigation

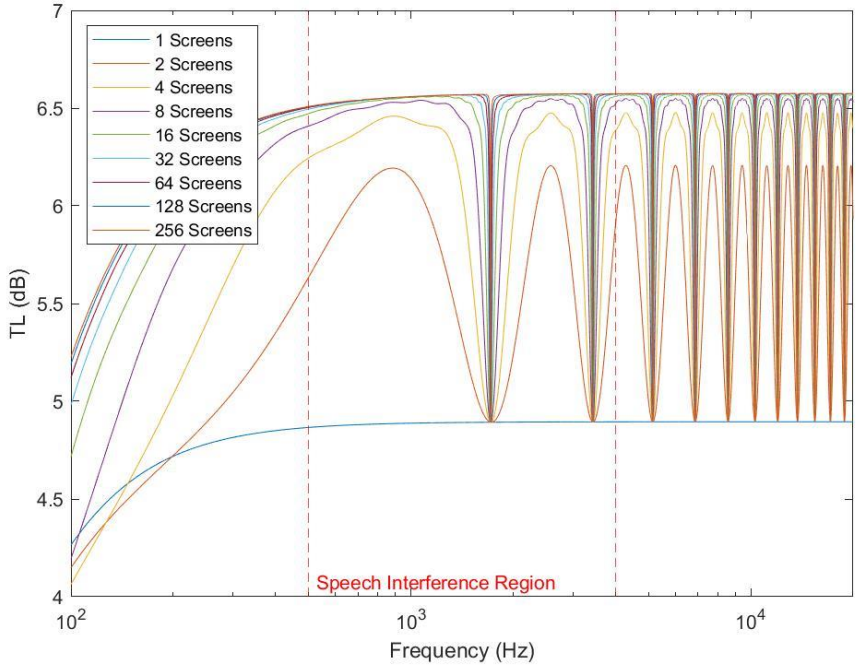
1 mm air gap between screens



Smaller air gaps:

- Same TL limits
- Depth resonances shift right, above 20kHz
- Transition now occurs in the SIR for n=256

10 cm air gap between screens

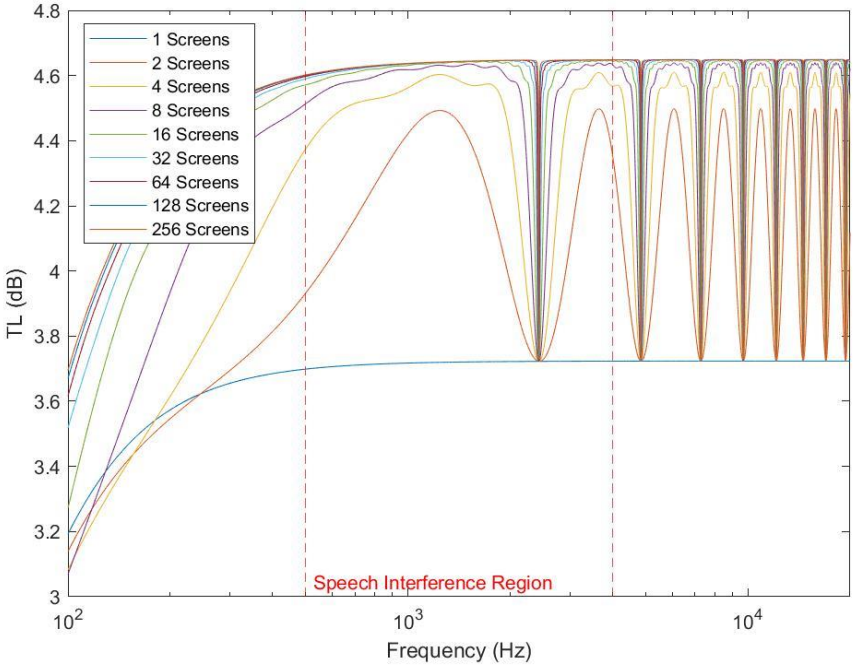


Larger air gaps:

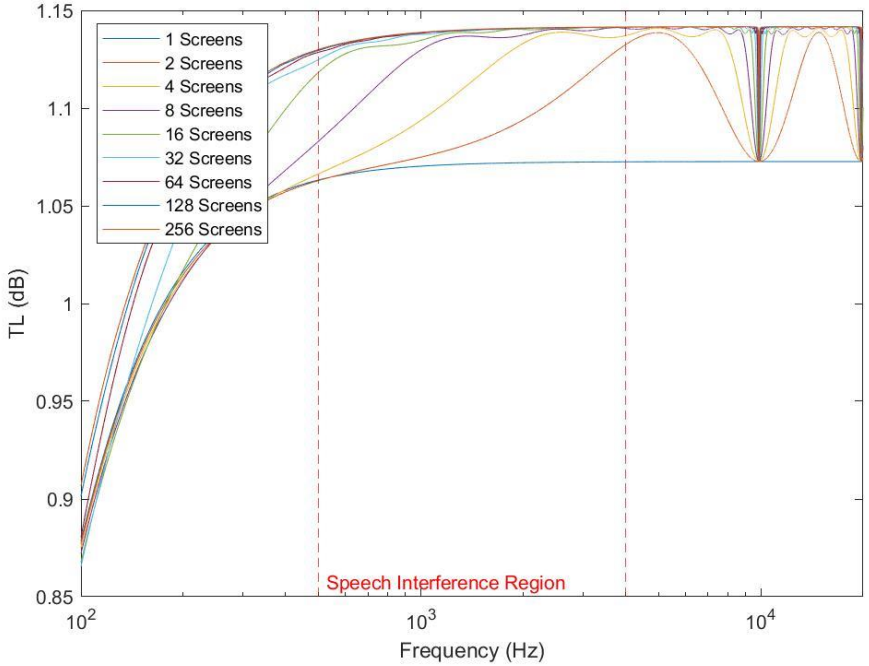
- Same TL limits
- Depth resonances shift left
- Depth resonances in the SIR for all n

Part 2 – Angle of Incidence Investigation

45° angle of incidence



80° angle of incidence



Increasing angle of incidence:

- Lower TL limits
- Depth resonances shift right
- Matches expectations:
  - k_z decreases so resonances shift up in frequency
  - z_{cn} increases so transmission coefficient approaches 1

THANK YOU

Questions?